Software Requirements Specification (SRS)

QuoteSpace — Quote Sharing Web Application

Version 1.0.0 | Release Date: October 2026

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1. Project Title and Purpose

Project Title: QuoteSpace (Configurable Application Name).
Purpose: QuoteSpace is a secure, modern, high-performance web platform designed to foster literary sharing, allowing visitors to discover inspiring quotes and registered users to publish, manage, and share their original or favorite literary quotes. The platform prioritizes high aesthetic presentation, mobile responsiveness, accessibility, and robust authorization.

2. Scope

The system encompasses a web-based client-server platform backed by Laravel and MySQL. Key functional areas include public quote browsing, real-time quote search, secure user registration and session management, owner-restricted CRUD operations for quotes, and custom error handling.

3. Intended Users & User Roles

• **Guest Visitor:** Unauthenticated user who can publicly browse quotes, search by content or author, paginate through results, and access registration/login pages.
• **Registered User:** Authenticated user who possesses full guest privileges plus the ability to create new quotes, view their personal 'My Quotes' dashboard, and edit or delete their own authored quotes with server-side policy enforcement.

4. Definitions and Terminology

• **CSRF:** Cross-Site Request Forgery security protection.
• **Policy:** Laravel server-side authorization class for resource ownership verification.
• **Eager Loading:** Eloquent query optimization technique avoiding N+1 performance bottlenecks.
• **Blade Inheritance:** Template layout architecture utilizing @extends and @yield.

5. Functional Requirements

Req ID	Module	Requirement Description	Validation / Constraint
FR-01	Auth	User Registration	Name: 2-50 chars, Email: unique & valid, Password: >=8 chars
FR-02	Auth	User Authentication & Session	Bcrypt password verify, session regeneration, POST logout
FR-03	Quotes	Public Quote Listing	Displays quote text, author display name, publication date
FR-04	Quotes	Quote Creation	Authenticated users only; content: 3-155 characters
FR-05	Quotes	Quote Editing & Update	Strict owner policy check; content: 3-155 characters
FR-06	Quotes	Quote Deletion	Strict owner policy check; POST/DELETE method spoofing + CSRF
FR-07	Search	Quote & Author Search	Filter by content or author; parameterized SQL queries; query preserved in pagination

FR-08	UI	Form Guidance & Feedback	Explicit hint text under every field; character counters; flash messages

6. Non-Functional Requirements

• **Performance Target:** Public quote listing page load time under 3.0 seconds under standard local/production server environment.
• **Usability Target:** First-time registered user quote creation completed in under 1 minute.
• **Security:** 100% server-side validation, CSRF protection on all forms, Bcrypt password hashing, parameterised SQL statements preventing SQL injection, strict Policy-based authorization preventing IDOR attacks.
• **Accessibility & UX:** Semantic HTML5 markup, WCAG AA color contrast, visible focus states, form input labels with explicit guidance hints beneath every field.

7. Business Rules & Validation Constraints

1. User Name must contain between 2 and 50 characters.
2. User Email must be valid and unique in the database.
3. User Password must contain at least 8 characters.
4. Quote Content must contain between 3 and 155 characters.
5. Ownership is strictly bound to auth()->id() on creation and verified via QuotePolicy prior to update or delete operations.

8. Testing Strategy & Execution Verification

The application includes 10 automated Feature/Unit test scenarios covering guest restrictions, authenticated creation, owner edit/delete permissions, unauthorized tampering blocks, input boundary validations, and public accessibility.